In [1]:
```
import numpy as np
```

In [2]:
```
from scipy import stats
```

In [3]:
```
marks = np.array([72, 68, 75, 70, 74, 69, 71, 73, 70, 72])
```

In [4]:
```
mu_0 = 70
```

In [5]:
```
t_stat, p_value = stats.ttest_1samp(marks, mu_0)
```

In [6]:
```
print(f"T-statistic: {t_stat:.3f}")
```

```
T-statistic: 1.993
```

In [7]:
```
print(f"P-value: {p_value:.4f}")
```

```
P-value: 0.0774
```

In [8]:
```
alpha = 0.05
```

In [9]:
```
if p_value < alpha:
    print("Reject Null Hypothesis → Mean is significantly different from 70.")
else:
    print("Fail to Reject Null Hypothesis → No significant difference.")
```

```
Fail to Reject Null Hypothesis → No significant difference.
```

In [11]:
```
import numpy as np
from math import sqrt
from scipy.stats import norm
# Given data
x_bar = 51.2 # sample mean
mu_0 = 50 # population mean
sigma = 3 # population standard deviation
n = 36 # sample size
# Calculate Z-statistic
z_stat = (x_bar - mu_0) / (sigma / sqrt(n))
# Two-tailed p-value
p_value = 2 * (1 - norm.cdf(abs(z_stat)))
print(f"Z-statistic: {z_stat:.3f}")
print(f"P-value: {p_value:.4f}")
alpha = 0.05
if p_value < alpha:
 print("Reject Null Hypothesis → Mean is significantly different from 50 g.")
else:
 print("Fail to Reject Null Hypothesis → No significant difference.")
```

```
Z-statistic: 2.400
P-value: 0.0164
Reject Null Hypothesis → Mean is significantly different from 50 g.
```

In [13]:
```
import numpy as np
from scipy import stats

# Data
```

```
A = [20, 22, 23]
B = [19, 20, 18]
C = [25, 27, 26]

# Perform one-way ANOVA
f_stat, p_value = stats.f_oneway(A, B, C)

print(f"F-statistic: {f_stat:.3f}")
print(f"P-value: {p_value:.4f}")

alpha = 0.05
if p_value < alpha:
 print("Reject Null Hypothesis → Means are significantly different.")
else:
 print("Fail to Reject Null Hypothesis → No significant difference.")
```

```
F-statistic: 25.923
P-value: 0.0011
Reject Null Hypothesis → Means are significantly different.
```

In [ ]: